Week 4 – End-to-End Data Analysis Case Study

Final Report

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Tools: Python, Pandas, Matplotlib, Scikit-learn, Power BI, Excel

1. Project Objective

The objective of this project was to perform an end-to-end analysis of sales data, identify important business patterns, evaluate sales and profitability, build a sales prediction model, and present actionable insights through a Power BI dashboard.

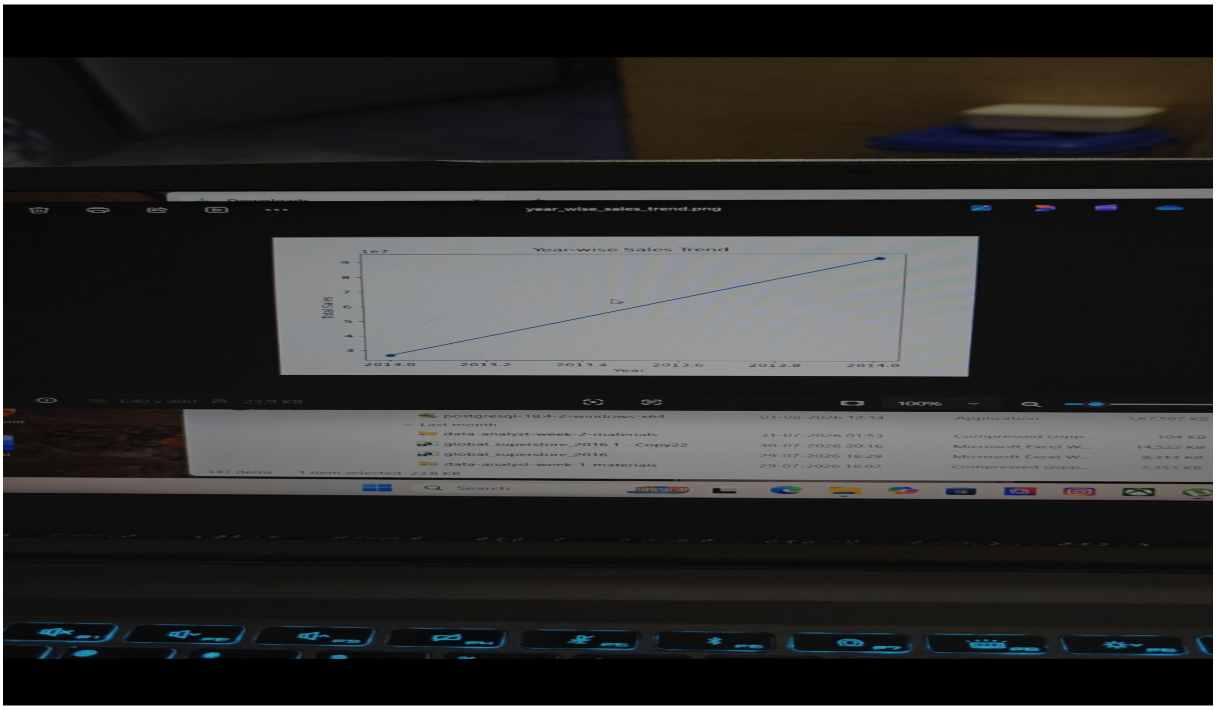
2. Dataset & Analysis Process

The dataset contains 700 records and business fields covering country, product, segment, units sold, sale price, discount, sales, cost of goods sold (COGS), profit, date, month and year. The workflow included data exploration and cleaning in Python, analysis using Pandas, visualization, regression modelling, and dashboard creation in Power BI.

3. Exploratory Data Analysis (EDA)

The EDA statistical summary showed 700 observations. Mean Units Sold was approximately 1,608.29, mean Manufacturing Price was approximately 96.48, mean Month Number was 7.90, and mean Year was 2013.75. The displayed missing-value check identified 53 missing values in Discount Band; the other displayed fields showed zero missing values.

Year-wise Sales Trend



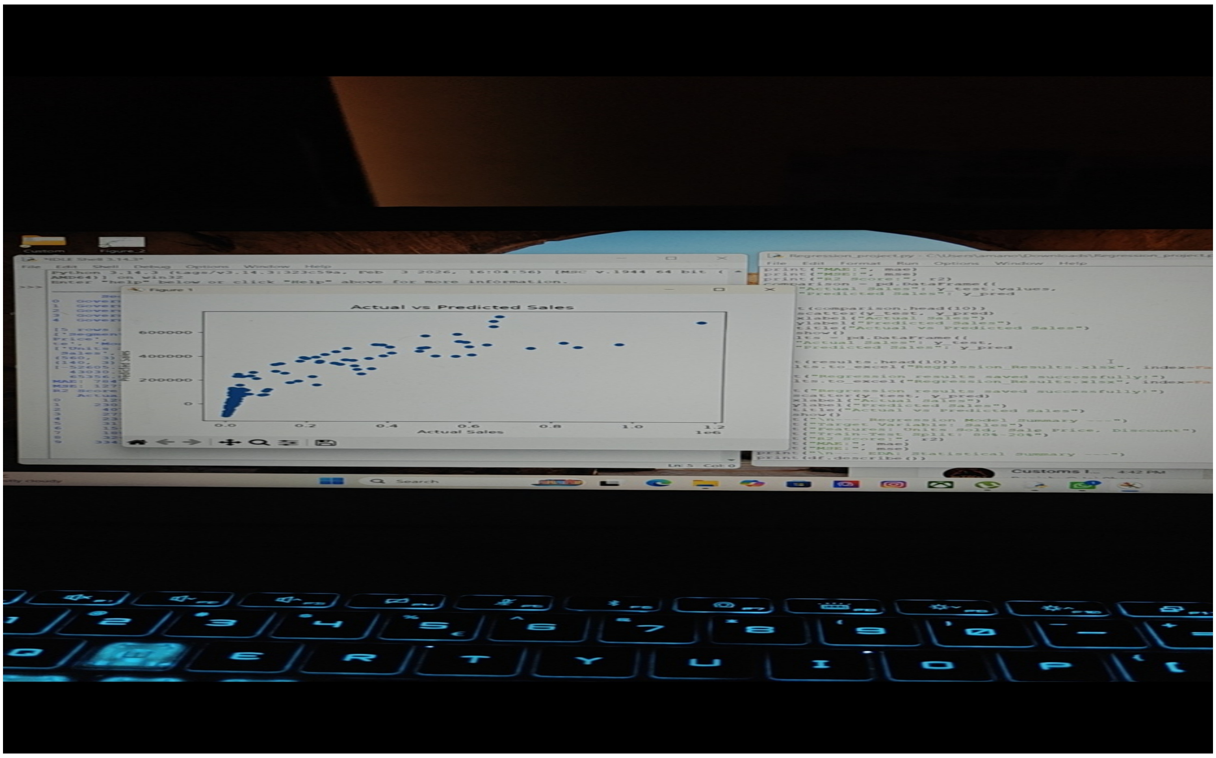
The year-wise chart shows a strong increase in sales from 2013 (approximately 26–27M) to 2014 (approximately 92–93M), indicating substantial year-over-year growth.

4. Regression Analysis

A regression model was developed with Sales as the target variable and Units Sold, Sale Price and Discount as predictors. An 80%–20% train-test split was used.

Metric	Result
R^2 Score	0.7698
MAE	78,476.14
MSE	1,275,048,953.82

The R^2 score of approximately 0.77 indicates that the selected features explain a substantial portion of the variation in Sales. The actual-versus-predicted plot also shows a clear positive relationship.

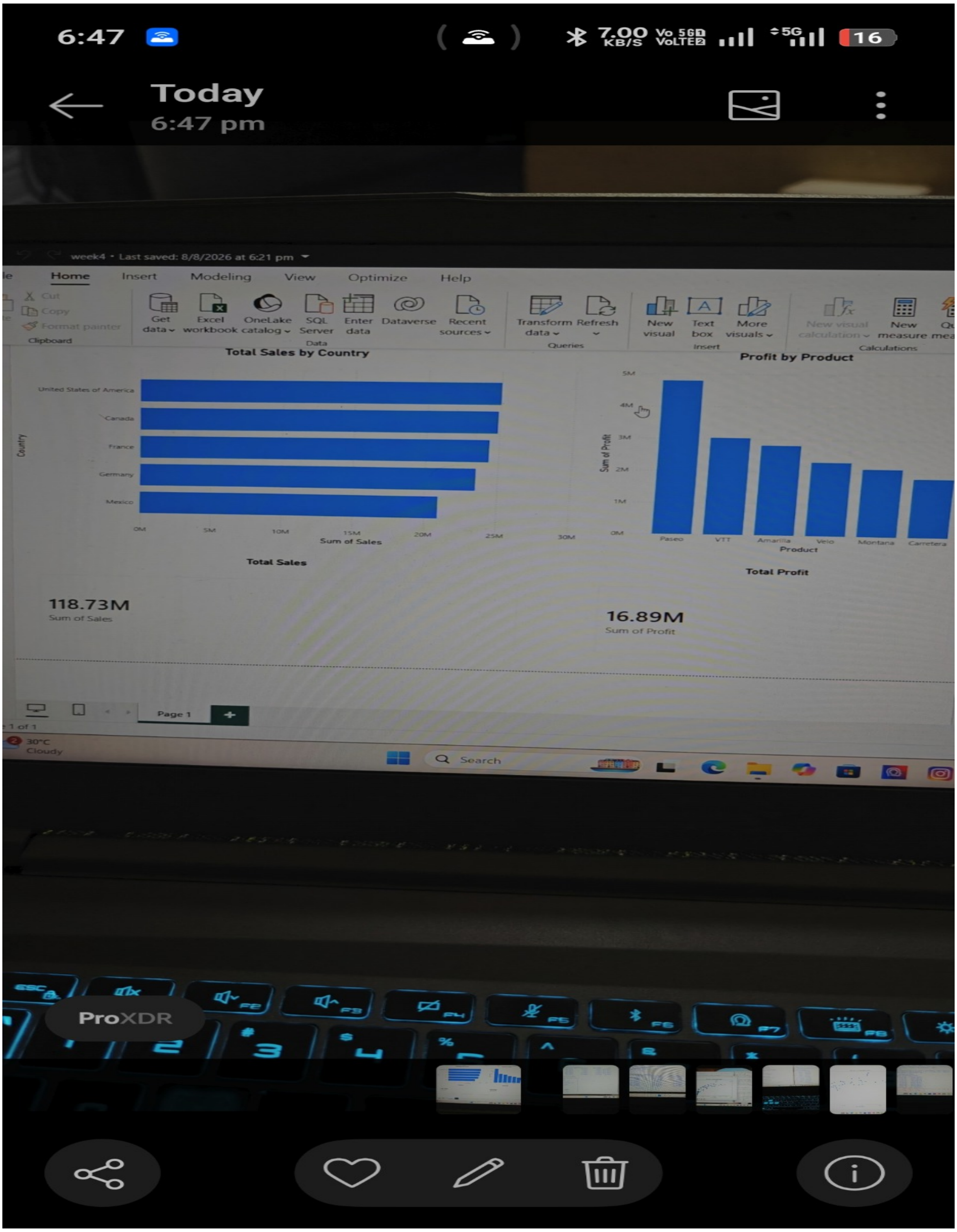


Actual vs Predicted Sales generated from the regression model.

5. Power BI Dashboard Findings

The Power BI dashboard summarizes the key business KPIs and compares sales by country and profit by product.

KPI / Finding	Result
Total Sales	118.73M
Total Profit	16.89M
Highest-sales country	United States of America
Highest-profit product	Paseo



Power BI dashboard showing country-wise sales, product-wise profit and overall KPIs.

6. Key Findings

• Total Sales are 118.73M and Total Profit is 16.89M.

• The United States of America is the highest-sales country in the dashboard.

• Paseo is the highest-profit product in the dashboard.

• Sales increased strongly from 2013 to 2014.

• The regression model achieved an R^2 score of 0.7698.

• The Enterprise segment showed negative profitability in the analysis and therefore needs attention.

7. Recommendations

1. Strengthen high-performing markets: Continue supporting the United States market while studying the practices behind its strong sales performance.

2. Promote high-profit products: Give greater marketing and sales focus to products such as Paseo that contribute strongly to profitability.

3. Investigate the Enterprise segment: Review pricing, discounts, COGS and operating costs to identify the reasons for negative profit and improve margins.

4. Optimize discounts: Evaluate whether discounts create enough incremental sales to justify their effect on profit.

5. Replicate successful growth drivers: Investigate the factors behind the large 2014 sales increase and apply successful strategies to future periods.

6. Monitor KPIs continuously: Use the Power BI dashboard to track sales, profit, country and product performance for data-driven decisions.

8. Final Conclusion

The project successfully combined EDA, predictive modelling and business intelligence visualization to turn raw sales data into actionable insights. The analysis identified major sales and profitability patterns, demonstrated a reasonably strong predictive relationship for Sales, and provided practical recommendations for improving business performance. The final Power BI dashboard can support ongoing monitoring and data-driven decision-making.

9. Deliverables Checklist

✓ Python data cleaning and EDA completed

✓ Regression analysis completed

✓ Power BI dashboard completed

✓ Key findings and recommendations prepared

✓ Final PDF explanation report prepared